CEFET/RJ

DAL & Comunidade

INTRODUÇÃO

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- R é uma linguagem de programação e ambiente de software livre para computação estatística
  - Apoiado pela *R Foundation for Statistical Computing*
  - Sistemas operacionais: Linux, Windows, Mac
- Criado por Ross Ihaka e Robert Gentleman na Universidade de Auckland, Nova Zelândia
- R foi derivado por S (Bell Laboratories - AT&T)
- R é uma linguagem amplamente usada por estatísticos, mineradores de dados e cientistas de dados
- Versão atual 4.2

Console R

RGui (64-bit) - [R Console]

Arquivo Editar Visualizar Misc Pacotes Janelas Ajuda

```
R version 4.2.1 (2022-06-23 ucrt) -- "Funny-Looking Kid"
Copyright (C) 2022 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64 (64-bit)

R é um software livre e vem sem GARANTIA ALGUMA.
Você pode redistribuí-lo sob certas circunstâncias.
Digite 'license()' ou 'licence()' para detalhes de distribuição.

R é um projeto colaborativo com muitos contribuidores.
Digite 'contributors()' para obter mais informações e
'citation()' para saber como citar o R ou pacotes do R em publicações.

Digite 'demo()' para demonstrações, 'help()' para o sistema on-line de ajuda,
ou 'help.start()' para abrir o sistema de ajuda em HTML no seu navegador.
Digite 'q()' para sair do R.

>
```

Disponível para Windows, Mac, Linux

R Console (operações básicas)

RGui (64-bit) - [R Console]

Arquivo Editar Visualizar Misc Pacotes Janelas Ajuda

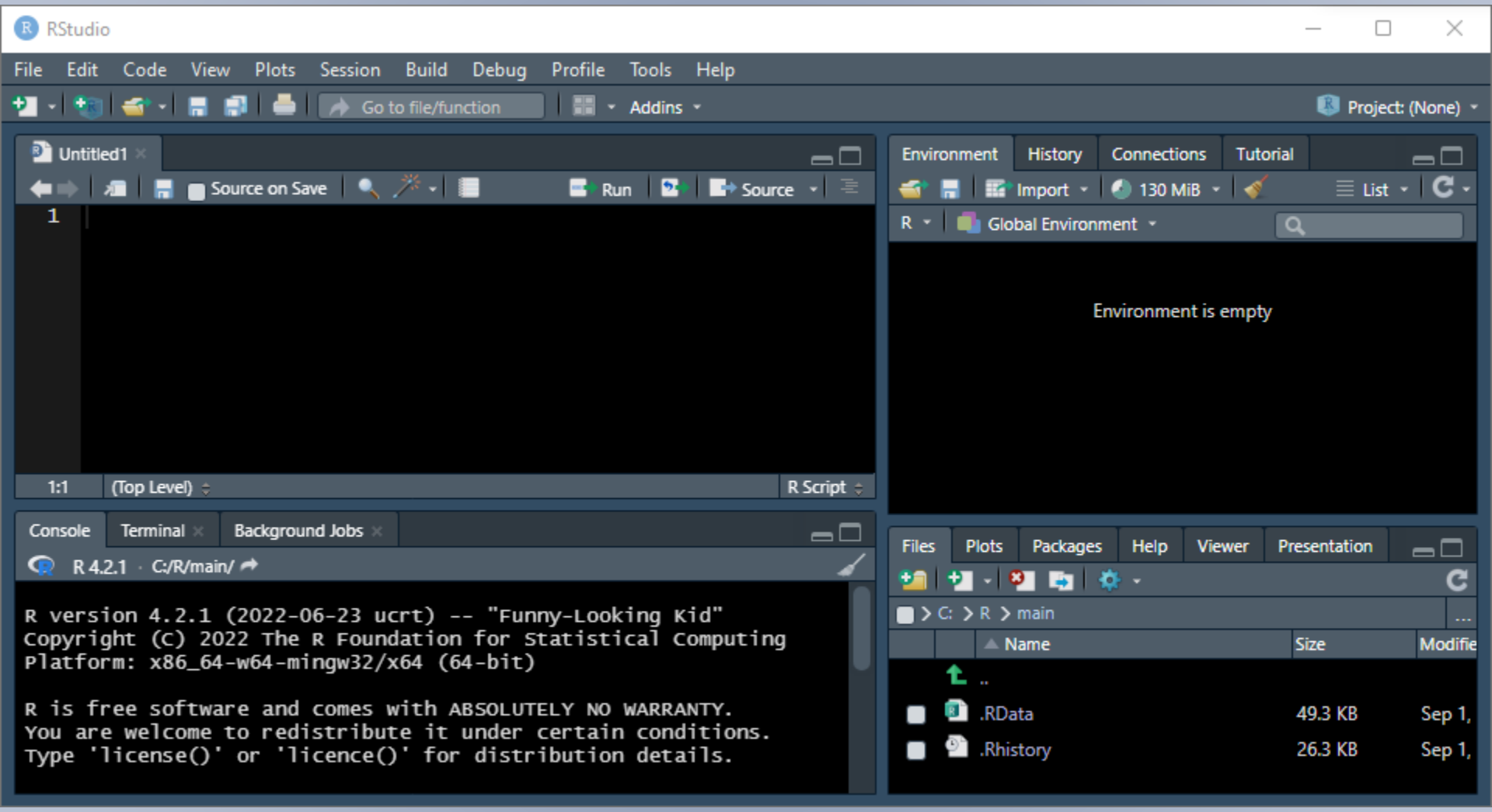
```
> 1 + 2
[1] 3
> 1:10
 [1]  1  2  3  4  5  6  7  8  9 10
> a <- 1
> a
[1] 1
> a + 2
[1] 3
> dado <- 1:6
> dado
[1] 1 2 3 4 5 6
> dado[1]
[1] 1
> dado[6]
[1] 6
> dado - 1
[1] 0 1 2 3 4 5
>
```

https://rstudio-education.github.io/hopr/basics.html

Instalação do R

- R
  - https://www.r-project.org
- Execute os exemplos anteriores

R Studio



Vantagens significativas: IDE com visualização de dados, depuração, criação de perfil

http://www.rstudio.com

Execução em bloco

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

01-primeiro-contato.R*

Source on Save | Run | Source

```
#parte 1
1 + 2
1:10
a <- 1
a
a + 2
dado <- 1:6
dado
dado[1]
dado[6]
dado - 1
```

6:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> 1 + 2
[1] 3
> 1:10
 [1]  1  2  3  4  5  6  7  8  9 10
> a <- 1
> a
[1] 1
>
```

Environment History Connections Tutorial

Import | 123 MiB | List

R | Global Environment

values

a	1

Files Plots Packages Help Viewer Presentation

C: > R > main

Name	Size	Modifie
..		
.RData	49.3 KB	Sep 1,
.Rhistory	26.3 KB	Sep 1,

Execução de trechos marcados no código (botão run)
Veja a janela de variáveis do ambiente

Execução completa

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

01-primeiro-contato.R

Source on Save | Run | Source

```
#parte 1
1 + 2
1:10
a <- 1
a
a + 2
dado <- 1:6
dado
dado[1]
dado[6]
dado - 1
```

12:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> source("C:/Users/eduar/OneDrive/Aulas/R/01-primeiro-contato.R")
>
```

Environment History Connections Tutorial

Import | 123 MiB | List

R | Global Environment

values

a	1
dado	int [1:6] 1 2 3 4 5 6

Files Plots Packages Help Viewer Presentation

C: > R > main

Name	Size	Modifie
..		
.RData	49.3 KB	Sep 1,
.Rhistory	26.3 KB	Sep 1,

Execução do código inteiro (botão source)

Instale o R Studio

- R Studio
  - https://www.rstudio.com
- Execute os exemplos anteriores
  - Código disponível em https://github.com/eogasawara/R

Operações aritméticas

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

01-primeiro-contato.R*

Source on Save | Run | Source

```
3 + 2
3 - 1
3 ^ 2
3 * 2
3 / 2
3 %% 2
```

17:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
[1] 5
> 3 - 1
[1] 2
> 3 ^ 2
[1] 9
> 3 * 2
[1] 6
> 3 / 2
[1] 1.5
> 3 %% 2
[1] 1
>
```

Environment History Connections Tutorial

Import | 123 MiB | List

R | Global Environment

Environment is empty

Files Plots Packages Help Viewer Presentation

C: > R > main

Name	Size	Modifie
..		
.RData	122 B	Sep 19
.Rhistory	26 KB	Sep 19

Operação com vetores

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

01-primeiro-contato.R*

Source on Save | Run | Source

```
2 * dado
dado * dado
```

27:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> 2 * dado
[1]  2  4  6  8 10 12
> dado * dado
[1]  1  4  9 16 25 36
>
```

Environment History Connections Tutorial

Import | 123 MiB | List

R | Global Environment

values

dado	int [1:6] 1 2 3 4 5 6

Files Plots Packages Help Viewer Presentation

C: > R > main

Name	Size	Modifie
..		
.RData	122 B	Sep 19
.Rhistory	26 KB	Sep 19

Funções

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

01-primeiro-contato.R*

Source on Save | Run | Source

```
round(3.1415)
round(3.1415, 2)
?round
mean(dado)
round(mean(dado))
```

33:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> round(3.1415)
[1] 3
> round(3.1415, 2)
[1] 3.14
> ?round
> mean(dado)
[1] 3.5
> round(mean(dado))
[1] 4
>
```

Environment History Connections Tutorial

Import | 131 MiB | List

R | Global Environment

Values

dado	int [1:6] 1 2 3 4 5 6

Files Plots Packages Help Viewer Presentation

R: Rounding of Numbers | Find in Topic

Round {base} R Documentation

Rounding of Numbers

Description

`ceiling` takes a single numeric argument `x` and returns a numeric vector containing the smallest integers not less

Argumento das funções

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

01-primeiro-contato.R

Source on Save | Run | Source

```
sample(x = dado, size = 2)
sample(dado, 2)
sample(x=dado, size=2, replace=TRUE)
?sample
args(sample)
```

42:1 (Top Level) R Script

Environment History Connections Tutorial

Import 157 MiB List

R Global Environment

Values

dado	int [1:6] 1 2 3 4 5 6

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> sample(x = dado, size = 2)
[1] 6 2
> sample(dado, 2)
[1] 3 6
> sample(x=dado, size=2, replace=TRUE)
[1] 1 5
> ?sample
> args(sample)
function (x, size, replace = FALSE, prob = NULL)
NULL
>
```

Files Plots Packages Help Viewer Presentation

R: Random Samples and Permutations | Find in Topic

sample {base} R Documentation

Random Samples and Permutations

Description

sample takes a sample of the specified size from the

Ordem dos parâmetros

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

01-primeiro-contato.R*

Source on Save | Run | Source

```
set.seed(1)
sample(x=dado, size=2, replace=TRUE)
sample(replace=TRUE, x=dado, size=2)
sample(dado, 2, TRUE)
```

49:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> set.seed(1)
> sample(x=dado, size=2, replace=TRUE)
[1] 1 4
> sample(replace=TRUE, x=dado, size=2)
[1] 1 2
> sample(dado, 2, TRUE)
[1] 5 3
>
```

Environment History Connections Tutorial

Import | 153 MiB | List

R | Global Environment

Values

dado	int [1:6] 1 2 3 4 5 6

Files Plots Packages Help Viewer Presentation

R: Random Samples and Permutations | Find in Topic

sample {base} R Documentation

Random Samples and Permutations

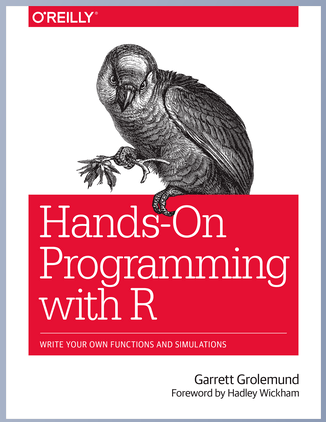
Description

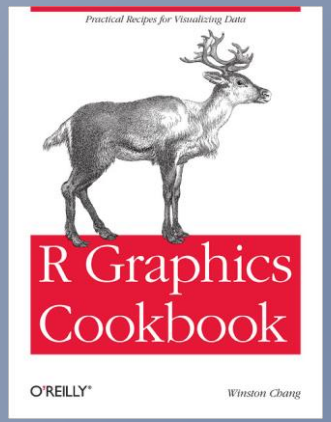
sample takes a sample of the specified size from the

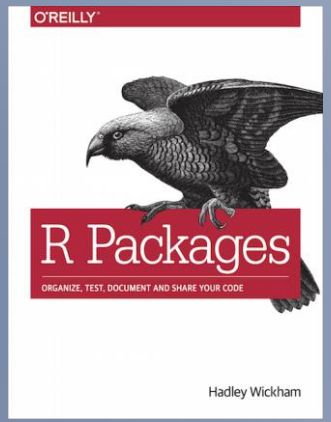
Chamadas com parâmetros explícitos não precisa seguir a ordem os argumentos

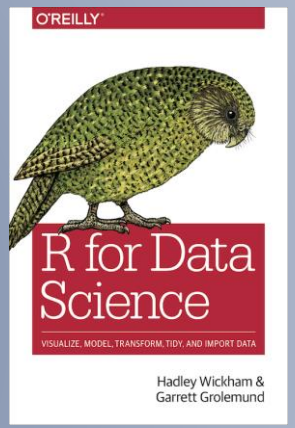
Referências

Código: https://nbviewer.org/github/eogasawara/R/blob/main/01-Introducao.ipynb









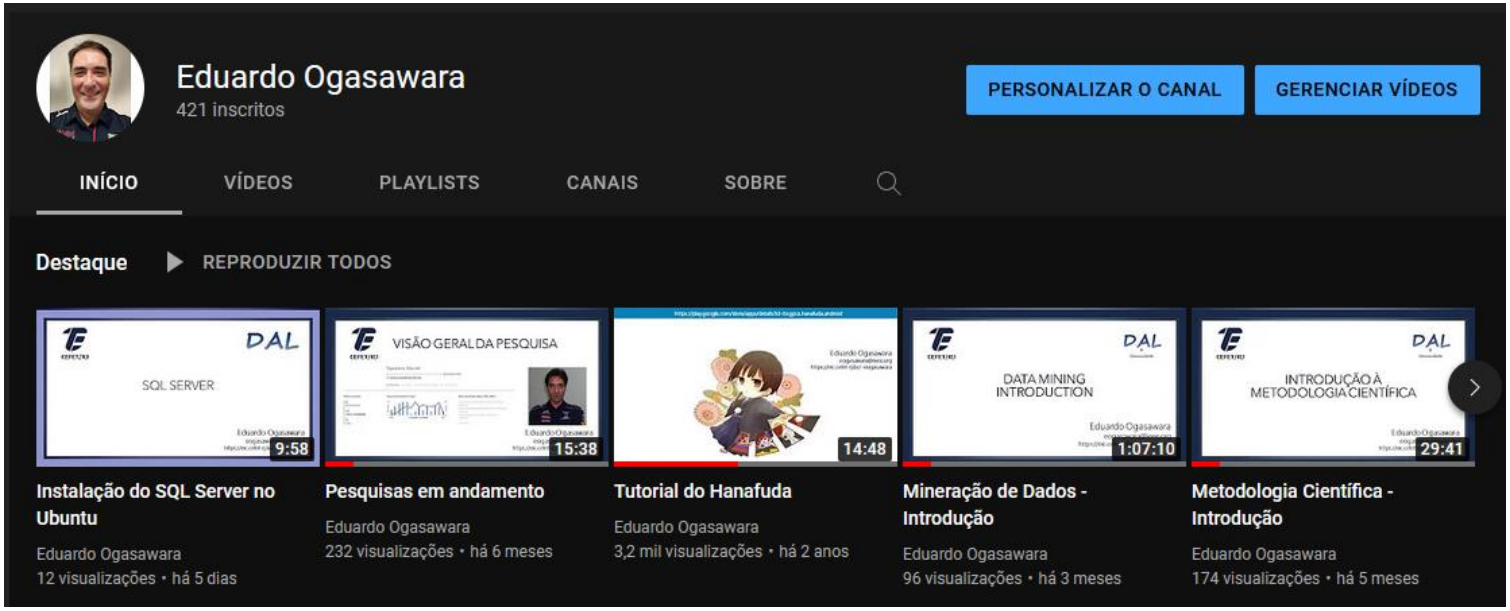
Hands-on Programming with R: https://rstudio-education.github.io/hopr/index.html
R Graphics Cookbook: https://r-graphics.org
R Packages: https://r-pkgs.org/index.html
R for Data Science: https://r4ds.had.co.nz

Leitura adicional: https://rstudio-education.github.io/hopr/basics.html

[1] G. Grolemund, 2014, Hands-On Programming with R: Write Your Own Functions and Simulations. O'Reilly Media, Inc.

Mantenha-se atualizado

Inscreva-se: https://eic.cefet-rj.br/~eogasawara/youtube



Slides and Jupyter notebooks: https://github.com/eogasawara/R